

Entropy Rigidity under Binary Refinement

and Hidden Correlation in FIN

Programme ST8547

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An analytical follow-up to ST8397–ST8546 with explicit proofs, a log-periodic counterexample, finite strict-graph computations, and independent numerical tests. The report distinguishes a conditional mathematical selection theorem from an experimental or parameter-free physical theory. It makes no claim of global mathematical priority.

Abstract

The logarithmic mean represents finite graph heat flow as the gradient flow of negative Shannon entropy, but the heat equation alone admits many entropy–mobility pairs. We characterize what an additional binary-refinement principle selects. In a common mass-coordinate, separable, edge-local class, equal binary splitting permits every curvature $\phi''(x) = g(\log_2 x)/x$ with positive continuous one-periodic g . An explicit smooth convex example disproves Shannon uniqueness under binary splitting alone.

Adding concavity of the diagonal mobility—in particular, joint convexity of the local flux cost—forces $\phi(x) = \kappa x \log x + \alpha x + \beta$. Equal binary and ternary compatibility is an alternative sufficient condition. Both are conditional theorems: metric compatibility is stronger than the heat intertwiner already used in FIN.

For the Shannon mobility we prove a positive coarse Onsager defect. On a connected graph with a strictly positive joint state, that defect vanishes exactly for a product state. A second exact identity writes the missing horizontal entropy dissipation as a weighted sum of conditional KL divergences. Thus a coarse heat equation may close perfectly while concealing fine correlations and dissipation. We exhibit this on the strict 12-to-24 FIN product graph.

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1 Evidence, conventions and priority

The mathematical inputs are the frozen strict conductances, the positive product refinement, and the entropy/Onsager formula in ST8397–ST8546. Project history and source boundaries were checked against AGENTS.md, the current-state compendium and the kernel-transformation diagram. This is a focused analytical follow-up using the repository state map, not a fresh verification of every historical report.

An audit of the previous 150 programme entries found 144 payload files equal to the empty JSON object. A successful hash replay confirms reproducibility of those files; it does not establish the mathematical assertions in their status labels. The present proofs and tests address the mathematics directly.

The logarithmic-mean heat-gradient identity and its divided-difference generalization are established results in finite Markov-chain transport. Maas [1] gives this framework, including homogeneity/concavity assumptions and the entropy-mobility relation. The results below are presented as checked consequences and a new connection within the FIN programme; world-first novelty has not been established.

Let $W = W^\top$, $W_{ij} \geq 0$, $W_{ii} = 0$, with connected support, and set

$$A = \text{diag}(W\mathbf{1}) - W.$$

For strict FIN, $n = 12$ and

$$W_{ij} = \frac{\cos(0.18575d + 0.16250)}{1 + d^{9/5}}, \quad d = \min(|i - j|, 12 - |i - j|).$$

All relevant cosine arguments for $d = 1, \dots, 6$ lie in $(0, \pi/2)$. The raw legacy kernel has signed weights and does not satisfy the same positive-conductance premise. No kernel completion or role transfer is made.

States are positive masses $p_i > 0$, with $\sum p_i = 1$ when probability is required. Scalar axioms are formulated on the full positive mass cone. The same scalar ϕ and the same dimensionless clock are used at every level; a reference measure is not silently renormalized.

2 Heat flow does not by itself select entropy

For $\phi \in C^2(0, \infty)$, $\phi'' > 0$, define

$$F_\phi(p) = \sum_i \phi(p_i), \quad (K_M^W(p)\xi)_i = \sum_j W_{ij} M(p_i, p_j) (\xi_i - \xi_j).$$

We require edgewise heat matching for every $a, b > 0$:

$$M(a, b) [\phi'(a) - \phi'(b)] = a - b. \tag{1}$$

This is a universal edge-local premise, not a conclusion from testing one state on a fixed complete graph.

Equation (1) gives

$$M_\phi(a, b) = \frac{a - b}{\phi'(a) - \phi'(b)}, \quad M_\phi(a, a) = \frac{1}{\phi''(a)}.$$

Then

$$K_{M_\phi}^W(p) \nabla F_\phi(p) = Ap.$$

For negative Shannon entropy, $\phi(x) = x \log x$,

$$\Lambda(a, b) = \frac{a - b}{\log a - \log b} = \int_0^1 a^t b^{1-t} dt, \quad \Lambda(a, a) = a. \tag{2}$$

The entropy dissipation is

$$\mathcal{I}_W(p) = \sum_{i < j} W_{ij}(p_i - p_j) \log(p_i/p_j) \geq 0. \quad (3)$$

Along heat flow $\dot{p} = -Ap$, negative Shannon F decreases and Shannon $S = -F$ increases. For boundary states, $K_p \log p$ must be interpreted through limiting edge fluxes; $0 \cdot \log 0$ in the gradient expression is not an ordinary finite product.

The identity represents a supplied heat generator. Microscopic rates are already contained in W ; they are not inferred by constructing K_p . Without edge locality, a positive matrix annihilating both $\mathbf{1}$ and $\log p$ can be added to K_p , leaving the vector field unchanged.

3 Refinement of masses and refinement of transport

On a finite product graph,

$$\tilde{A} = A \otimes I_m + I_n \otimes L_B, \quad C = I_n \otimes \mathbf{1}_m^\top.$$

C sums masses and is different from the Hilbert isometry $R = I_n \otimes \mathbf{1}_m / \sqrt{m}$. For a product state $P_{ib} = p_i r_b$, $r_b > 0$, $\sum_b r_b = 1$, impose

$$CK_M^{\tilde{W}}(P)C^\top = K_M^W(p). \quad (4)$$

Lifted potentials are constant in each fiber; vertical edges disappear from the left side. Consequently the edgewise condition is

$$\sum_b M(r_b a, r_b b) = M(a, b). \quad (5)$$

Equal binary splitting gives

$$2M(a/2, b/2) = M(a, b). \quad (6)$$

The repository's linear heat intertwiner $C\tilde{A} = AC$ holds for every fine state and every vertical rate. It does not imply (4). The latter preserves a quadratic form on *all* potentials, not just the entropy gradient used in the heat equation.

4 A complete counterfamily for binary uniqueness

Theorem 1 (Dyadic classification). *Under (1), equal binary compatibility (6) holds if and only if*

$$\phi''(x) = \frac{g(\log_2 x)}{x},$$

where g is positive, continuous and one-periodic.

Proof. Take $a = b = x$ in (6). The continuous diagonal formula implies $\phi''(x/2) = 2\phi''(x)$. Thus $x\phi''(x)$ is invariant under doubling. Conversely, integrate this curvature from $b/2$ to $a/2$. A change of variable gives $\phi'(a/2) - \phi'(b/2) = \phi'(a) - \phi'(b)$, which proves (6). $\square$

For $\omega = 2\pi/\log 2$ and $0 < |\varepsilon| < 1$, take

$$\phi_\varepsilon(x) = x \log x - \frac{\varepsilon x}{\omega(1 + \omega^2)} [\cos(\omega \log x) + \omega \sin(\omega \log x)].$$

Then

$$\phi_\varepsilon''(x) = \frac{1 + \varepsilon \sin(\omega \log x)}{x} > 0.$$

This is a smooth convex entropy on positive masses, continuous at zero. Its mobility generates exactly the same heat flow, and it passes every equal dyadic subdivision. It is not Shannon plus an affine function.

At $\varepsilon = 1/4$, $a = 0.07$, $b = 0.19$, the binary relative defect is 2.22×10^{-16} , whereas ternary and 0.3/0.7 splitting defects are 0.09275176 and 0.08029200. Special pairs with dyadic ratios may conceal this difference. All numeric figures are diagnostics for the analytic counterexample.

5 Conditional entropy rigidity

Theorem 2 (Binary refinement and convex transport). *Assume (1), (6), and that the diagonal mobility $f(x) = M(x, x)$ is concave. Then*

$$\boxed{\phi(x) = \kappa x \log x + \alpha x + \beta, \quad M(a, b) = \Lambda(a, b)/\kappa, \quad \kappa > 0.}$$

Proof. The dyadic condition gives $f(x/2) = f(x)/2$. The periodic classification and positive continuity bound $f(x)/x$, so $f(0+) = 0$. For any concave f with $f(0) = 0$, $f(x)/x$ is nonincreasing. But this ratio takes equal values at x and $2x$, hence is constant on $[x, 2x]$. Those intervals cover all positive scales. Write $f(x) = x/\kappa$; then $\phi''(x) = \kappa/x$. Integrating proves the entropy formula, and (1) determines the mobility. $\square$

Diagonal concavity follows from joint convexity of the local transport cost

$$\Psi(a, b, j) = \frac{j^2}{2M(a, b)}.$$

To see this, restrict to $a = b = x$, and at endpoints choose $j_i = f(x_i)$. At weight t , put $s = tf(x_1) + (1-t)f(x_2) > 0$. Convexity gives $s^2/[2f(tx_1 + (1-t)x_2)] \leq s/2$, hence concavity of f . Conversely Λ is jointly concave by (2), and the convex perspective $j^2/(2z)$, decreasing in $z > 0$, proves convexity of the Shannon transport cost.

This is a sufficient mathematical principle coupling refinement, information and transport. The cost-convexity and full metric-compatibility premises are additional assumptions for FIN. The result does not derive W , its frozen parameters or the time unit.

Removed premise	Counterexample or remaining freedom
Diagonal concavity	The log-periodic family survives all binary splits; its diagonal mobility has $f''(1) > 8$ at $\varepsilon = 1/4$.
Binary compatibility	Quadratic entropy with $M = 1$ has convex transport cost and heat matching, but the binary left side is 2.
Heat matching	$M = \sqrt{ab}$ is homogeneous and concave; it does not force a separately supplied entropy to generate heat.
Specified mass coordinates	Transporting a reference measure permits non-Shannon relative-entropy counterexamples.

Theorem 3 (Two independent subdivisions). *Heat matching plus continuous compatibility with equal binary and ternary subdivision also forces the Shannon/logarithmic-mean pair up to scale.*

Proof. $h(x) = x\phi''(x)$ is invariant under multiplying by 2 and by 3. On logarithmic coordinates the periods are $\log 2$ and $\log 3$. Their ratio is irrational, since $2^m = 3^n$ has no positive integer solution. Their integer span is dense; continuity makes h constant. $\square$

Arbitrary binary split fractions similarly imply additive $s \mapsto M(sa, sb)$, hence degree-one homogeneity and Shannon. These conditions are stronger than equal binary subdivision alone.

5.1 Reference-measure caveat

For a reversible reference π , use density $u_i = p_i/\pi_i$ and relative entropy $\sum_i \pi_i \phi(u_i)$. If π is simultaneously refined to $\pi \otimes r$, a product state $p \otimes r$ keeps density u_i unchanged in each child. Any convex ϕ can then inherit product mobility compatibility. Quadratic relative entropy is an explicit non-Shannon example. The preceding rigidity theorem is not invariant under arbitrarily changing its mass/reference-measure conventions.

6 Hidden correlation despite exact coarse dynamics

Let $P_{ib} > 0$, $p_i = \sum_b P_{ib}$, and $r_i(b) = P_{ib}/p_i$. For Shannon mobility define

$$\mathcal{D}(P) = K_\Lambda^W(p) - CK_\Lambda^{\tilde{W}}(P)C^\top.$$

Theorem 4 (Coarse Onsager defect). $\mathcal{D}(P)$ is positive semidefinite. On a connected base graph it vanishes if and only if $P = p \otimes r$.

Proof. Its edge conductance is $W_{ij}\delta_{ij}$, where

$$\delta_{ij} = \Lambda(p_i, p_j) - \sum_b \Lambda(P_{ib}, P_{jb}) \geq 0.$$

Using the integral in (2), Hölder's inequality proves this bound. For $0 < t < 1$, equality holds exactly when the normalized child distributions are equal: $r_i = r_j$. Integrating retains the strict equality case. Thus the defect is a positive Laplacian, zero exactly when equality holds on every base edge. Connectedness gives independence globally. $\square$

Theorem 5 (Hidden horizontal entropy production). With $\mathcal{I}_W$ as in (3),

$$\sum_b \mathcal{I}_W(P_b) - \mathcal{I}_W(p) = \sum_{i < j} W_{ij} [p_i D_{\text{KL}}(r_i \| r_j) + p_j D_{\text{KL}}(r_j \| r_i)] \geq 0.$$

Proof. On each edge split $\log(P_{ib}/P_{jb}) = \log(p_i/p_j) + \log(r_i(b)/r_j(b))$. The first part sums to the coarse dissipation; the two remaining weighted terms are exactly the two KL divergences. $\square$

The equality condition is the same independence criterion. Full fine entropy production additionally contains the vertical contribution. The defect is independent of that vertical rate. It requires fine-state information and cannot be reconstructed from a coarse-only heat record.

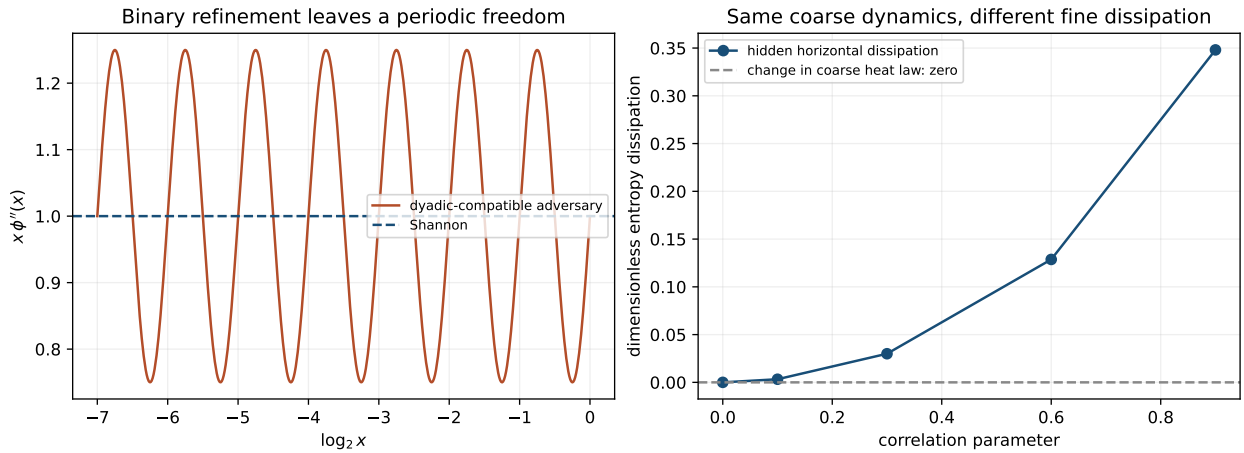


Figure 1: The dyadic-compatible non-Shannon entropy (left) and the hidden horizontal dissipation under exactly unchanged coarse heat dynamics (right). Both plots come from the executable calculations.

7 Finite strict-graph verification

The package contains 15 independent tests of the formulas and counterexamples, including high-precision integration of the logarithmic mean and differentiation of the non-Shannon entropy. It tests 80 positive strict-graph states, 12 dyadic depths, biased and ternary splits, correlated product refinements, the KL identity, the coarse heat semigroup and unitary spectral preservation.

Computed check	Value
Largest heat identity error, 80 states	6.67×10^{-16}
Largest product Onsager defect	3.79×10^{-16}
Correlated example, coarse heat error	1.39×10^{-17}
Hidden horizontal dissipation	0.1286853406
KL identity error	3.06×10^{-16}
Smallest positive defect eigenvalue	0.0006361160
Largest defect eigenvalue	0.0034115110

For the correlated example, the defect has eleven positive eigenvalues and one mass-conservation zero mode. A second example has uniform coarse probabilities: coarse heat remains stationary while hidden dissipation is strictly positive. No laboratory data or interval certification is claimed.

8 Physical meaning, limitations and next research

The significant result for FIN is the coupling of entropy and mobility by a precisely stated refinement/convexity principle, together with an exact fine/coarse correlation diagnostic. The theorems are valid on any connected positive weighted graph. They are not a fingerprint specific to the strict parameter tuple.

The same A still supplies e^{-itA} , e^{-tA} and $\cos(t\sqrt{A})$. Unitary evolution preserves the density-matrix spectrum, whereas the heat channel dissipates the convex entropy. Refinement rigidity constrains the heat representation, not the measurement rule, the microscopic configuration generator or an absolute clock.

Neither the new convexity axiom nor the stronger mobility equation has been derived from the nadsoliton's historical coupling diagram. The factor κ , graph parameters, reference measure convention, physical preparation, units, continuum tail and apparatus remain outside the conclusions.

The earlier abstract no-equivariant-section lemma is valid only when its claimed vertical group preserves the admissible class. Arbitrary tail translations do not generally preserve positivity or a physical action. They must not be imported as physical symmetries without proof.

The following tasks have the highest decision value:

1. Derive or refute joint convexity of the transport cost from an independent FIN action, without choosing it to obtain Shannon.
2. Determine whether an actual coarse measurement requires full Onsager compatibility or only the weaker heat intertwiner.
3. Extend the correlation defect and its equality conditions through multiple refinement levels.
4. Prove quantitative stability bounds for approximate dyadic compatibility and approximate convexity.
5. Construct a physical preparation/measurement map only after the preceding source premises are explicit.

The log-periodic counterexample rules out automatic Shannon selection by binary self-similarity. The conditional rigidity theorem explains one sufficient extra principle. The KL identity identifies information hidden by a coarse layer. None establishes a physical Theory of Everything.

A Reproduction

From the package directory:

```
python3 analysis.py --output results.json
python3 -m unittest -v test_analysis.py
MPLCONFIGDIR=/tmp/fin-st8547-mpl python3 plot_results.py
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

Dependencies are Python with NumPy, SciPy, mpmath and Matplotlib, plus a standard pdfLaTeX installation for the PDF. Exact dependency versions used are recorded in provenance.json. Proofs are also supplied as PROOFS_EN.md. No long-running computation or network access is required for replay.

References

- [1] J. Maas, *Gradient flows of the entropy for finite Markov chains*, Journal of Functional Analysis 261 (2011), 2250–2292. <https://arxiv.org/abs/1102.5238>.
- [2] FIN local source: ST8397–ST8546, logarithmic-mean mobility; ST30, ST245 and ST1137, product refinement; AGENTS.md and the current-state compendium. Their source hashes and the audit scope are listed in provenance.json.